QUICKRESERVE SYSTEM

FINAL PROJECT PRESENTATION

IT 112 – Web Systems and Technologies

Submitted to:

Dr. Guillermo V. Red, Jr., DIT

GROUP MEMBERS

NAME	ROLE	MAIN RESPONSIBILITIES
Nathaniel Raynada	Project Manager	Overall team coordination, milestone tracking, communication, and project supervision
Nathaniel Raynada	Front-End Developer	Designs and develops HTML, CSS, Bootstrap, and JavaScript user interface components
Anna Mae Broma	Back-End Developer	Implements Node.js server, Express routes, backend logic, and API integration
Camille Bercilla	GitHub Manager	Handles repository management, commits, version control, and collaboration workflows
Justine Dasallas	Database Manager	Design and manages MongoDB schema, integrates mongoose
Mariel Odoño	Documentation Officer	Creates reports, technical documentation, and project files
John Edmar Bongcodin	Tester/Debugger	Testing and Debugging

PROJECT INTRODUCTION

QuickReserve is a web-based bus ticket reservation system that allows users to easily search available trips, select seats, and book tickets online. It provides a faster and more convenient alternative to traditional booking methods by allowing users to access the system anytime using a web browser. The system also helps manage reservations efficiently by storing data in a secure database and providing organized records for both users and administrators.

PROJECT OBJECTIVE

The main objectives of the QuickReserve system are:

- To develop a web-based bus ticket reservation system
- To allow users to easily search available trips and schedules
- To provide a simple and convenient seat booking process
- To store and manage booking records efficiently using a database
- To reduce manual booking and improve transaction speed
- To improve user experience through a user-friendly interface

TECHNOLOGY STACK

Frontend

- HTML – used to structure the web pages
- CSS – used for design and styling of the system
- JavaScript – adds interactivity and dynamic features
- Bootstrap – makes the design responsive and mobile-friendly

Backend

- Node.js – server-side environment for backend operations
- Express.js – handles routing and API requests

Database

- MongoDB Atlas – cloud database used for storing user, reservation, and booking data

Version Control / Collaboration

GitHub – used for version control and collaboration during the development of the QuickReservE System

SYSTEM FEATURE OVERVIEW

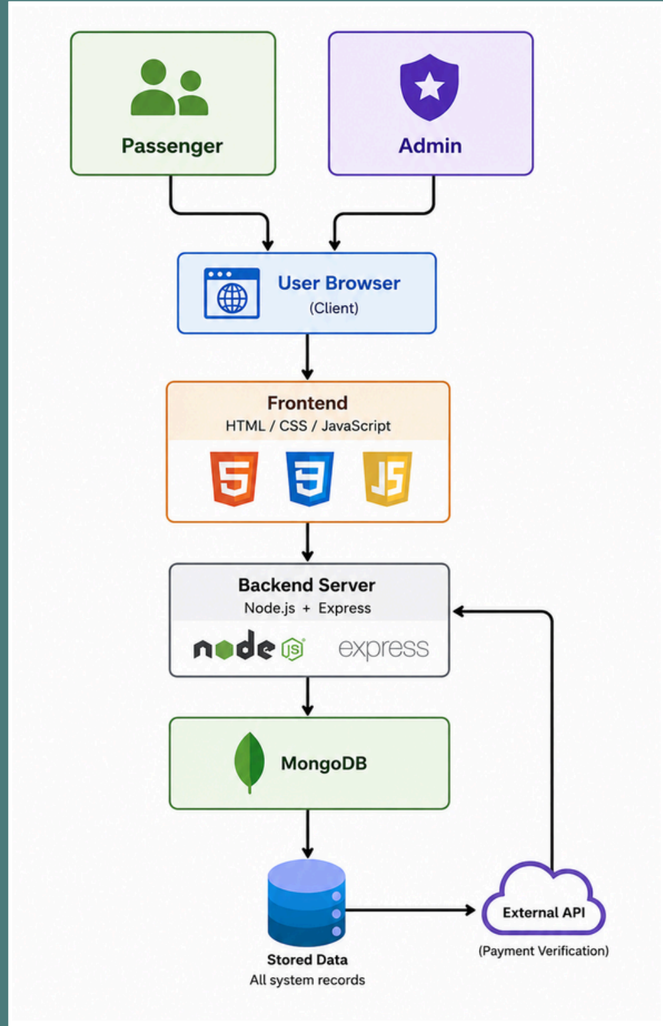
User Features

- Register and log in to the system
- Search available bus trips and schedules
- View trip details and seat availability
- Select preferred seats
- Book tickets online
- View, manage, or cancel reservations

Admin Features

- Add, update, and manage bus trips
- Control seat availability per trip
- View all user bookings and reservations
- Monitor system activity and records
- Ensure data is updated and organized

SYSTEM ARCHITECTURE



Passengers and admins access the QuickReservE System using a web browser. The frontend was developed using HTML, CSS, JavaScript, and Bootstrap, while the backend uses Node.js and Express.js. All data such as passenger information, bus schedules, bookings, and payments are stored in MongoDB Atlas.

LIVE DEMONSTRATION

CHALLENGES ENCOUNTERED

- Difficulty connecting the frontend and backend during development Errors encountered while integrating the API and database
- Problems handling and displaying booking data correctly
- Adjusting the user interface to improve responsiveness and usability
- Managing time and coordinating tasks among group members
- Debugging system errors during testing and implementation

How we handled the challenges to solve these challenges, the group worked together by dividing tasks and helping each member troubleshoot errors during development. We conducted testing regularly to identify problems in the system and applied necessary fixes to improve functionality and performance. We also used online resources, documentation, and team collaboration to better understand technical issues and complete the project successfully.

FUTURE IMPROVEMENTS

- Add an online payment system for more convenient transactions
- Improve the user interface and overall system design
- Add email or SMS notifications for booking confirmation
- Enhance system security and user data protection
- Improve mobile responsiveness for better accessibility on different devices
- Add real-time seat availability and trip updates
- Implement advanced search and filtering features for users

THANK YOU!